Predictive Analytics for Smart City Infrastructure: Multi-Horizon Traffic Congestion Forecasting and Explainable Agentic Signal Advisory Using Spatiotemporal Transformers and RAG

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***Abstract*—Urban traffic management needs forecasts that are accurate at several horizons, honest about their uncertainty, and explainable enough to justify an intervention. This paper presents an end-to-end research prototype for Manhattan that couples a context-gated spatiotemporal graph transformer for 15/30/60-minute link-speed forecasting with a constrained, retrieval-augmented agentic workflow that produces simulation-only signal-timing advisories. The forecaster consumes a directed road-link graph built from NYC DOT real-time speed links as a PyTorch Geometric Temporal signal, fuses weather, 311 and calendar context through an availability-aware gate, and is calibrated with split conformal prediction. Explanations are computed on the trained model with grouped KernelSHAP. Advisories are drafted by a locally hosted large language model that must cite passages retrieved from an allowlisted corpus of federal signal-timing guidance and a project-defined policy, are checked by deterministic safety constraints, and are evaluated against a baseline controller in SUMO with matched seeds before being queued for human review. All data are public and all reported figures are generated from stored artifacts; results, including negative and blocked outcomes, are reported in Section VII exactly as produced by the pipeline.**

***Index Terms*—traffic forecasting, graph neural networks, transformers, conformal prediction, explainable AI, retrieval-augmented generation, agentic systems, traffic simulation, smart cities**

I. Introduction

Urban traffic-management centres typically observe congestion as it happens. Linking an *advance* forecast to an explainable, evidence-grounded and tested operational recommendation remains difficult: traffic depends on recurrent demand, neighbouring links, weather, planned events, disruptions and infrastructure condition, and a forecast alone does not establish which intervention helps. An LLM-generated signal recommendation is, by itself, not evidence that an action is safe or effective.

We study a decision-support pipeline for the scenario of a permitted event near a busy corridor in which rain and a nearby collision coincide. The pipeline (i) forecasts link speeds 15, 30 and 60 minutes ahead with a context-gated graph spatiotemporal Transformer, (ii) attaches split-conformal prediction intervals, (iii) explains the forecast of the actual model with grouped Shapley values, (iv) retrieves operational guidance with a FAISS index, (v) asks a local LLM (Qwen2.5-7B-Instruct via Ollama) for schema-constrained candidate actions, (vi) applies deterministic safety constraints, and (vii) tests admissible candidates in a SUMO microscopic simulation before human review.

Proposed contributions (stated as proposals; their effects are reported only where measured): (1) availability-aware context fusion in the graph forecaster; (2) an explicit uncertainty-and-evidence gate that decides whether advisory generation proceeds; (3) a simulation-based candidate gate with side-road protection; and (4) a traceable evaluation that links forecast quality, grounding, deferral and simulated intervention outcomes to versioned artifacts. We do not claim deployment, real-world congestion reduction or causal explanations.

II. Related Work

Graph traffic forecasting. Diffusion-convolutional recurrent networks [1], spatio-temporal graph convolution [2] and Graph WaveNet [3] established graph neural networks for sensor-network speed forecasting; attention-based variants use graph attention [4] and Transformer encoders [5]. Our model combines GATv2 message passing with a temporal Transformer and adds an availability-aware gate over feature groups; we treat it as a project-designed architecture, not a new model family.

Uncertainty. Split conformal prediction [6], [7] gives finite-sample marginal coverage under exchangeability; for time series this assumption can fail, motivating adaptive and non-exchangeable variants [8], [9]. We therefore report empirical coverage by horizon and scenario.

Explainability. SHAP [10] attributes predictions to inputs; for tree baselines we use TreeSHAP, and for the graph model a grouped KernelSHAP over feature groups and graph neighbourhoods.

Retrieval-augmented LLMs. RAG [11] conditions generation on retrieved passages; we embed passages with Sentence-BERT models [12] and search them with FAISS [13]. The LLM is Qwen2.5 [14].

Simulation. SUMO [15] with TraCI is widely used to evaluate signal control. We use it as a scenario-based testbed and do not claim calibration to real demand.

III. Problem Formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a directed graph whose nodes are monitored directed road links and whose edges connect a link

to links that can receive its traffic. At each 5-minute step t every node v carries a feature vector $x_{t,v}$ partitioned into groups $g \in \{$traffic, calendar, weather, infrastructure, events, incidents$\}$ with availability indicators $m_{t,v,g}$. Given the last $L = 24$ steps, the task is to predict observed link speed $y_{t+h,v}$ for $h \in \{3, 6, 12\}$ steps (15/30/60 min), together with an interval $[\hat{y} - q_h, \hat{y} + q_h]$. A downstream advisory is admissible only if data quality, interval width, retrieved evidence, deterministic constraints and a simulation comparison all pass; otherwise the system returns a distinct DEFER or REJECT status. Training minimises a masked Huber loss over observed targets only.

IV. Data

All sources are public (Table I). Speed targets come from the NYC DOT Real-Time Traffic Speed Data feed (Socrata `i4gi-tjb9`), restricted to Manhattan links with geometry, for 2026-01-01 to 2026-10-20. Each record carries a floating local timestamp that is localised to `America/New_York` and converted to UTC; the ambiguous daylight-saving hour is nulled rather than guessed. Records are snapped to a right-labelled 5-minute grid per link, averaged within a bin, bounded to physically plausible speeds, and forward-filled causally for at most three steps. Context comes from Open-Meteo (historical forecasts for the operational track, the finalised archive for the retrospective track), NYC 311 service requests with a declared 24-hour lag, Motor Vehicle Collisions (retrospective only; the feed lagged publication by about four months at download time) and NYC permitted events. Sampled automated traffic volume counts parameterise simulation demand only. OpenStreetMap supplies geometry for the simulated corridor. Every download is logged with URL, time, HTTP status and SHA-256, and a server-side audit of dataset identity and observed date ranges precedes every download; the full-year link enumeration timed out server-side, so the link set is the 125 links active since 2026-09-20, which is recorded as a limitation.

V. Proposed Methodology

A. Context-gated GATv2–Temporal Transformer

For each node and time step, every feature group g is projected to a common width $d = 64$, $h_{t,v,g} = \mathrm{ReLU}(W_g x_{t,v,g})$. A scoring network gives $s_{t,v,g} = w^\top \tanh(U h_{t,v,g})$; scores of unavailable groups ($m_{t,v,g} = 0$) are masked to $-\infty$ and a softmax over the remaining groups yields weights $\alpha_{t,v,g}$. The fused representation is $z_{t,v} = \sum_g \alpha_{t,v,g} h_{t,v,g}$; if no group is available a learned "unavailable" vector is used and the event is logged. Two GATv2 layers [4] (4 heads, residual connections, layer normalisation) propagate z over the fixed directed graph at every time step (implemented with PyTorch Geometric [16]). Each node's sequence then receives a learned positional embedding and passes through a 2-layer, 4-head Transformer encoder (feed-forward width 128, dropout 0.1); a linear head on the last state outputs the three horizons. Targets are standardised per link with training-period statistics and the masked Huber loss is averaged over observed targets only; batches without any observed target are skipped and counted. Optimisation: AdamW (lr 10^{-3}, weight decay 10^{-4}), batch size 8, gradient clipping at 1.0, early stopping on validation MAE with patience 5 and at most 30 epochs, seed 42. The temporal snapshots are held in a PyTorch Geometric Temporal [17] `StaticGraphTemporalSignal`, from which training windows are drawn lazily. The gate weights are not interpreted as causal importance.

B. Baselines and ablations

Persistence (last observation), a training-period historical average per link, day type and 5-minute slot, XGBoost and LightGBM (one model per horizon on 12 speed lags and the same causal features), a GRU, and a temporal-only Transformer (the proposed model without message passing). Ablations remove the gate (concatenation instead), or one feature group at a time.

C. Uncertainty and reliability gate

With the selected model frozen, per-horizon absolute residuals on the calibration partition give the split-conformal quantile q_h as the $\lceil (n+1)(1-\alpha) \rceil$-th smallest residual ($\alpha = 0.1$); a per-link variant is also computed when at least 50 calibration residuals exist. The advisory gate defers when data are older than 20 minutes, more than one third of recent traffic inputs are missing, the interval is wider than a threshold chosen on calibration data (80th percentile of per-link widths), or calibration support is below 200 samples.

D. Explanations

For an alert (link v, horizon h, origin t) the explained scalar is the predicted speed in mph. Players are the target's recent traffic (last 30 min), older traffic (30–120 min), the traffic of its graph neighbours and each context group; an absent player takes values from 16 training-period background windows. All 2^P coalitions are evaluated (exact KernelSHAP [10]). We additionally measure neighbour-edge removal and temporal-window sensitivities. Tree baselines are explained with TreeSHAP and labelled as baseline explanations.

E. Retrieval, LLM and agentic workflow

Approved documents (U.S. federal signal-timing guidance, SUMO documentation, and a clearly labelled project-defined experimental policy) are hashed, allow-listed, chunked to at most 200 word-pieces (below the 256-token limit of all-MiniLM-L6-v2) and indexed with a FAISS inner-product index over normalised embeddings. A query whose best passages score below 0.35 returns INSUFFICIENT_EVIDENCE. The LLM receives forecasts, intervals, quality flags, computed attributions, retrieved passages (marked as untrusted text) and the allowed action catalogue, and must return JSON that satisfies a Pydantic schema enforced through Ollama's structured-output option. A LangGraph state machine runs quality $\rightarrow$ forecast $\rightarrow$ uncertainty $\rightarrow$ explanation $\rightarrow$ retrieval $\rightarrow$ recommendation $\rightarrow$ safety $\rightarrow$ simulation $\rightarrow$ decision $\rightarrow$ human review, with deterministic routing to DEFER_DATA,

DEFER_UNCERTAINTY, DEFER_EVIDENCE, REJECT_SCHEMA, REJECT_CONSTRAINT, REJECT_SIMULATION, PENDING_HUMAN_REVIEW or ERROR_TOOL. Only the recommendation node calls the LLM.

F. Safety constraints and simulation gate

Allowed actions are a bounded green extension (0–10 s) of an existing green phase, selection of a pre-validated program, or monitoring only. Thirteen deterministic checks (mapping validity, signal and phase existence, action type, bounds, minimum/maximum green, clearance preservation, pedestrian and side-road protection availability, data freshness, forecast reliability, evidence support) plus a text policy filter must pass. Candidates are then applied through a protective TraCI wrapper that refuses non-green phases and out-of-bound durations, and compared with the fixed-time baseline on matched seeds; a candidate passes only if corridor delay improves by at least 1%, side-road delay rises by at most 10% and no vehicle teleports. Pedestrian performance is not modelled and is reported as NOT_EVALUATED.

VI. Experimental Design

Partitions. The 5-minute time axis is split chronologically into training (60%), validation (15%), calibration (10%) and test (15%). A forecast origin belongs to a partition only if all of its targets fall inside it (purging); earlier history may be used as input. Scalers and link eligibility (training coverage ≥ 0.6) use the training partition only.

Feature tracks. The *operational* track uses traffic, calendar, archived weather *forecasts* (Open-Meteo historical-forecast API; issue times are not exposed, a disclosed limitation) and 311 requests with a declared 24-hour publication lag. The *retrospective* track replaces forecasts by reanalysis weather and adds collisions and borough-level permitted events, whose operational availability cannot be established; it is used for contextual analysis only.

Metrics. MAE, RMSE, sMAPE and R^2 per horizon on a shared valid-target mask; paired day-block bootstrap for model differences; empirical interval coverage and width by horizon, congestion category and scenario; Recall@k, MRR and support-status accuracy for retrieval; schema validity, invalid-citation counts, constraint violations and decision distribution for advisories; SUMO time loss, travel time, waiting time, stops, queue length, throughput, side-road delay, unfinished trips, insertion backlog and teleports; and latency of each stage.

Advisory cases and policies. Routine rush hour, rainy period (if present in the test partition), a stale-data stress test, a wide-interval stress test, an out-of-scope query (insufficient evidence), and alerts on links with a reviewed mapping to the simulated corridor. Policies: full system; LLM without retrieval; RAG without the constraint layer; RAG with constraints but without the simulation gate.

TABLE I
VERIFIED PUBLIC DATA SOURCES (SERVER-SIDE AUDIT)

NOT RUN

TABLE II
TEST-PARTITION FORECAST ERROR (MPH) BY MODEL AND HORIZON, OPERATIONAL FEATURE TRACK

NOT RUN

VII. Results

Every table in this section is generated by `python -m smartcity.cli paper sync` from the result files in `reports/experiments/`; a table reading NOT RUN means the corresponding stage has not produced an artifact and no number is claimed for it. The narrative below is completed only from those artifacts.

A. Data coverage and sources

B. Multi-horizon forecasting

C. Uncertainty calibration

D. Evidence retrieval

E. Advisory workflow

F. Simulation testbed

Status. The real-data run is in progress on the authors' workstation; the forecasting, calibration, retrieval, advisory and simulation tables above are regenerated when it completes, and this narrative is written from them. No quantitative claim is made in this version of the manuscript.

VIII. Discussion

The system is designed so that each component's contribution can be isolated: baselines and the proposed model share one test mask; ablations remove one context group or the gate at a time; the advisory evaluation runs the same cases under four policies that switch off retrieval, the constraint layer or the simulation gate. The discussion of what these comparisons show is written from the generated tables once the real-data run has completed and is therefore absent from this version rather than anticipated. What can already be stated is architectural: the language model never holds safety authority (constraints and the simulation gate are deterministic code), retrieved text is treated as untrusted evidence that must be cited by chunk id, and the most permissive status the workflow can reach is approval for simulation, never deployment.

IX. Limitations

The DOT link set is sparse and highway-heavy, so graph neighbours are often distant and the spatial layer has less to exploit than on a dense sensor network; the full-year link enumeration timed out server-side and the study uses the 125 links active since 2026-09-20. Operational weather is Open-Meteo's historical-forecast product, a re-forecast proxy rather than an archived agency forecast. Collisions lag publication

TABLE III
ABLATIONS OF THE PROPOSED MODEL (TEST MAE, MPH, MEAN OVER HORIZONS)

NOT RUN

TABLE IV
SPLIT-CONFORMAL INTERVALS: EMPIRICAL TEST COVERAGE (NOMINAL 90%) AND WIDTH

NOT RUN

TABLE V
RETRIEVAL EVALUATION ON THE LABELLED QUERY SET

NOT RUN

TABLE VI
ADVISORY WORKFLOW OUTCOMES BY POLICY (REPLAY CASES; SIMULATION-ONLY EFFECTS)

NOT RUN

by months and are retrospective only. A single seed and a single chronological split are used; the only variance estimate is a day-block bootstrap on paired differences. Split conformal intervals carry a marginal guarantee under exchangeability, which traffic series violate. The approved corpus is small and the retrieval labels were drafted by the project without external review. The language model is a 7B quantised model served locally; its advisories are only as good as the evidence and the constraints that follow it. The SUMO testbed is a scenario built from OSM geometry with partly assumed demand and default signal programs, not a calibrated digital twin, and contains no pedestrian demand. Container deployment was not exercised end to end because of registry rate limits in the development environment. The complete, maintained list is in `reports/limitations.md`.

X. CONCLUSION AND FUTURE WORK

We presented a complete, reproducible prototype that joins multi-horizon graph-transformer forecasting, conformal calibration, model-faithful explanations, allowlisted retrieval, a locally hosted language model constrained by deterministic checks, and matched-seed simulation into one auditable advisory workflow on public New York City data. Conclusions about accuracy, calibration and advisory quality are drawn only from the generated tables and are added when the real-data run completes. Future work includes multi-seed evaluation, a denser sensor network, calibrated simulation demand from continuous counts, pedestrian modelling, and an externally reviewed retrieval benchmark.

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TABLE VII
SUMO TESTBED (SIMULATION): MEAN OVER COMPLETED MATCHED SEEDS

NOT RUN